THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

LAB ASSIGNMENT-3

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2O23

1.Write a program to implement:
(a) Pointer to an object (b) this pointer Practice both . (dot operator) and -> (arrow operator).

A3_1.cpp

```cpp
#include <iostream>
using namespace std;
class Demo {
    int x;
public:
    void set(int x) {
    this->x = x;
    }
    void show() {
    cout << "Value of x = " << x << endl;
    }
};
int main() {
    Demo d;
    Demo *ptr = &d;
    ptr->set(10);
    d.show();
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Value of x = 10

--------------------------------
Process exited after 0.2912 seconds with return value 0
Press any key to continue . . .
```

2. Write a program to swap private values of two classes using a friend function.

A3_1.cpp A3_2.cpp

```cpp
#include <iostream>
using namespace std;
class B;
class A {
    int x;
public:
    void set(int a) { x = a; }
    void show() { cout << "A.x = " << x << endl; }
    friend void swap(A &, B &);
};
class B {
    int y;
public:
    void set(int b) { y = b; }
    void show() { cout << "B.y = " << y << endl; }
    friend void swap(A &, B &);
};
void swap(A &a, B &b) {
    int temp = a.x;
    a.x = b.y;
    b.y = temp;
}

int main() {
    A a;
    B b;
    a.set(5);
    b.set(10);
    cout << "Before swapping:\n";
    a.show();
    b.show();
    swap(a, b);
    cout << "\nAfter swapping:\n";
    a.show();
    b.show();
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Before swapping:
A.x = 5
B.y = 10

After swapping:
A.x = 10
B.y = 5

--------------------------------
Process exited after 0.2446 seconds with return value 0
Press any key to continue . . .
```

3. Write a program to add data objects of two different classes using friend functions.

A3_1.cpp A3_2.cpp A3_3.cpp

```cpp
#include <iostream>
using namespace std;
class B;
class A {
    int x;
public:
    A(int a) { x = a; }
    friend int add(A, B);
};
class B {
    int y;
public:
    B(int b) { y = b; }
    friend int add(A, B);
};
int add(A a, B b) {
    return a.x + b.y;
}
int main() {
    A a(10);
    B b(20);
    cout << "Sum = " << add(a, b);
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Sum = 30
--------------------------------
Process exited after 0.2465 seconds with return value 0
Press any key to continue . . .
```

4. Write a program to demonstrate the working of a friend class.

A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp

```cpp
#include <iostream>
using namespace std;
class First {
    int x = 10;
    friend class Second;
};
class Second {
public:
    void show(First f) {
    cout << "Value of x = " << f.x;
    }
};
int main() {
    First f;
    Second s;
    s.show(f);
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Value of x = 10
--------------------------------
Process exited after 0.26 seconds with return value 0
Press any key to continue . . .
```

5. Write a program using Array of Objects to display the area of multiple rectangles.

A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp A3_5.cpp

```cpp
#include <iostream>
using namespace std;
class Rectangle {
    int l, b;
public:
    void set() {
    cin >> l >> b;
    }
    void area() {
    cout << "Area = " << l * b << endl;
    }
};
int main() {
    Rectangle r[3];
    for (int i = 0; i < 3; i++) {
    cout << "Enter length and breadth: ";
    r[i].set();
    }
    for (int i = 0; i < 3; i++) {
    r[i].area();
    }
    return 0;
}
```

OUTPUT:

C:\Users\DELL\Desktop\OOPS

```
Enter length and breadth: 45
1
Enter length and breadth: 45
2
Enter length and breadth: 45
3
Area = 45
Area = 90
Area = 135

--------------------------------
Process exited after 12.19 seconds with return value 0
Press any key to continue . . .
```

6. Write a program to define function cube() as inline for calculating cube of a number.

A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp A3_5.cpp A3_6.cpp

```cpp
#include <iostream>
using namespace std;
inline int cube(int x) {
    return x * x * x;
}
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    cout << "Cube = " << cube(n);
    return 0;
}
```

OUTPUT:

C:\Users\DELL\Desktop\OOPS

```
Enter a number: 4
Cube = 64
--------------------------------
Process exited after 3.349 seconds with return value 0
Press any key to continue . . .
```

7. Write a program to pass an object as an argument and return the object from a function.(a) Use pass-by-value (b) Use pass-by-address.

A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp A3_5.cpp A3_6.cpp A3_7a.cpp

```cpp
#include <iostream>
using namespace std;
class Test {
public:
    int x;
};
Test change(Test t) {
    t.x = 20;
    return t;
}
int main() {
    Test t1, t2;
    t1.x = 10;
    t2 = change(t1);
    cout << "Returned value = " << t2.x;
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Returned value = 20
--------------------------------
Process exited after 0.2769 seconds with return value 0
Press any key to continue . . .
```

A3_1.cpp | A3_2.cpp | A3_3.cpp | A3_4.cpp | A3_5.cpp | A3_6.cpp | A3_7a.cpp | A3_7b.cpp

```cpp
#include <iostream>
using namespace std;
class Test {
public:
    int x;
};
Test change(Test *t) {
    t->x = 30;
    return *t;
}
int main() {
    Test t1, t2;
    t1.x = 10;
    t2 = change(&t1);
    cout << "Returned value = " << t2.x;
    return 0;
}
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS

Returned value = 30
--------------------------------
Process exited after 0.2881 seconds with return value 0
Press any key to continue . . .
```